

Qn	Ans	Solution
7	D	Weight of boat = upthrust = weight of fresh water displaced = 35.6 kN Weight of boat = weight of salt water displaced = $mg = \rho Vg = 35.6 \text{ kN}$ $V = \frac{35.6 \times 10^3}{\rho g} = \frac{35.6 \times 10^3}{(1024)(9.81)}$ $= 3.54 \text{ m}^3$
8	D	5.45